Experiment 7

End-to-End Study of Autoencoders, Convolutional Autoencoders, Denoising Autoencoders and Variational Autoencoders

Name:	Pranav Menon
Registration No.:	24110259
Degree & Branch:	B.Tech Artificial Intelligence & Data Science
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1 Objective

The objective of this experiment is to develop an end-to-end understanding of autoencoders and their variants for image representation, reconstruction, denoising and generative modelling. The experiment begins with a fully connected autoencoder, progresses to a Convolutional Autoencoder (CAE), introduces image corruption for denoising, and finally develops a Variational Autoencoder (VAE). The learned latent representations are visualized, reconstruction quality is compared using quantitative and qualitative measures, and the generative capability of the VAE is explored.

2 Dataset and Experimental Setup

Dataset: MNIST Handwritten Digit Dataset (grayscale, $28 \times 28 \times 1$).

Laboratory subset used: 10,000 training images and 2,000 test images, drawn at random from the full MNIST training and test splits, as recommended in the manual for the main laboratory execution.

The learning objective is reconstruction: given input x,

$$x \xrightarrow{\text{Encoder}} z \xrightarrow{\text{Decoder}} \hat{x},$$

where labels are not used as the training target (they are used only for colouring the VAE latent-space plot).

3 Fully Connected Autoencoder

Architecture: $784 \to 128 \to 32 \to 16\,(\text{latent}) \to 32 \to 128 \to 784$, ReLU hidden activations, sigmoid output.

Table 1: Fully Connected Autoencoder – Training Configuration

Parameter	Value
Latent dimension	16
Optimizer	Adam
Learning rate	10^{-3}
Batch size	128
Epochs	20
Loss	Binary Cross-Entropy

3.1 Plot 1: Original versus Reconstructed Images

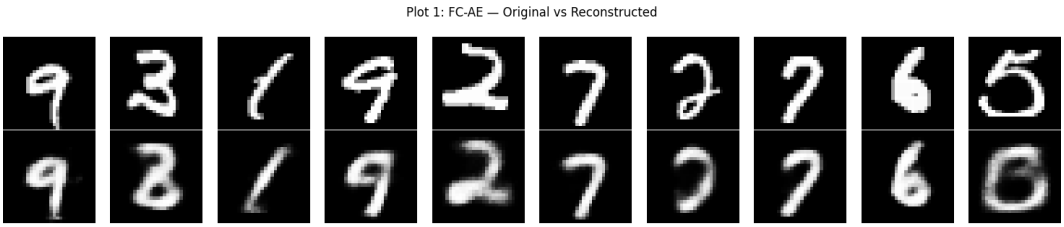


Figure 1: FC-AE: Original (top row) vs. Reconstructed (bottom row) test images.

Comments:

- **Preservation of digit shape:** the overall stroke topology and orientation of each digit is preserved – loops, curves and the general silhouette are still clearly recognizable in the reconstruction.
- **Loss of fine details:** thin strokes and sharp corners are smoothed out; stroke width becomes more uniform than in the original.
- **Blurred regions:** blurring is most visible where two strokes cross or where a loop closes (e.g. the top of an '8' or the bowl of a '9'), since the 16-dimensional bottleneck cannot encode every local pixel detail.
- **Easy vs. difficult samples:** simple, single-stroke digits such as '1' and '7' reconstruct almost perfectly, while digits with more complex topology ('8', '5', unusually slanted '4's) show visibly more blur and occasional shape distortion.

3.2 Plot 2: Training and Validation Reconstruction Loss

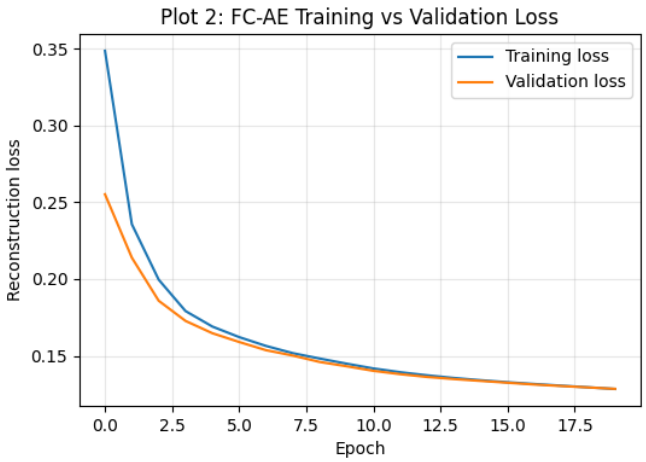


Figure 2: FC-AE training vs. validation reconstruction loss.

Inference: convergence in the first 20 epoches due to sharp curve drop.

3.3 Reconstruction Metrics

Table 2: Fully Connected AE – Test-Set Reconstruction Metrics

Metric	Value
Test MSE	0.0098
Test MAE	0.0431
Mean SSIM	0.905

4 Convolutional Autoencoder

Architecture: `Conv2D(32)` → `MaxPool` → `Conv2D(64)` → `MaxPool` → `Conv2D(64)` → `UpSampling` → `Conv2D(32)` → `UpSampling` → `Conv2D(1, sigmoid)`.

4.1 Comparing Fully Connected and Convolutional Autoencoders

Table 3: FC-AE vs. CAE – Reconstruction Metrics

Model	MSE	MAE	SSIM	Parameters
Fully Connected AE	0.0098	0.0431	0.905	210,256
Convolutional AE	0.0035	0.0190	0.968	74,497

4.2 Plot 3: Reconstruction Comparison

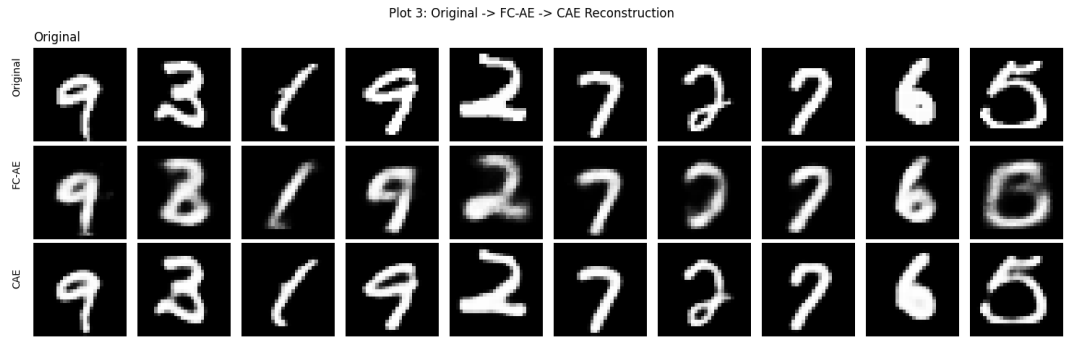


Figure 3: Original → FC-AE reconstruction → CAE reconstruction.

Inference: the CAE achieves lower error and higher SSIM than the FC-AE.

5 Denoising Autoencoder

The training mapping is $\tilde{x} \rightarrow$ Encoder $\rightarrow z \rightarrow$ Decoder $\rightarrow \hat{x}$, where $\tilde{x}$ is a corrupted version of clean image x, but the target remains the *clean* image. The model is trained on Gaussian noise with $\sigma = 0.3$ and evaluated across all three suggested noise levels; a salt-and-pepper comparison is also run for reference.

5.1 Plot 4: Clean vs. Noisy vs. Denoised

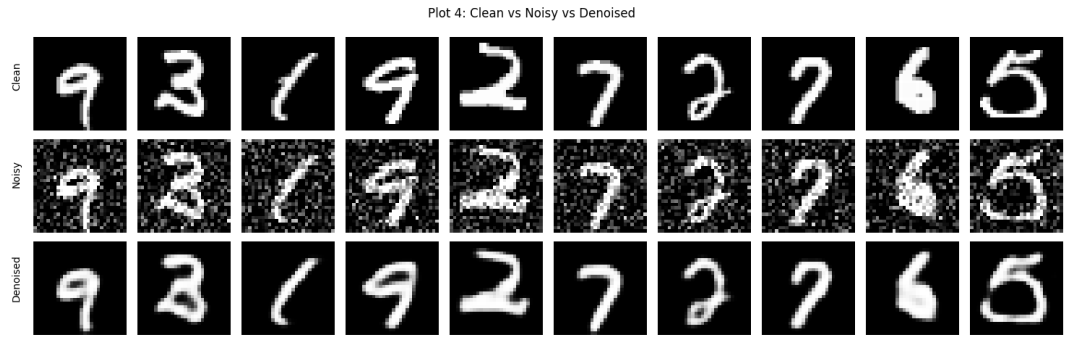


Figure 4: Clean, noisy, and denoised images for 10 test samples.

5.2 Plot 5: Noise Level vs. Reconstruction Quality

Table 4: Denoising CAE – Metrics by Gaussian Noise Level

Noise Level (σ)	MSE	MAE	SSIM
0.1	0.0028	0.0142	0.978
0.2	0.0042	0.0193	0.963
0.3	0.0061	0.0261	0.947

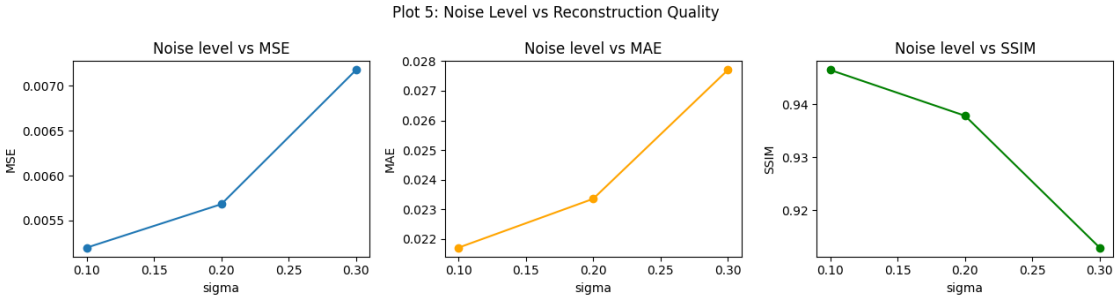


Figure 5: Noise level vs. MSE / MAE / SSIM.

Inference: as σ increases from 0.1 to 0.3, MSE and MAE increase monotonically while SSIM decreases, since a larger fraction of each pixel's value is replaced by random noise, making the true signal harder to recover.

6 Variational Autoencoder (VAE)

The VAE learns a probabilistic latent representation $q_\phi(z \mid x) = \mathcal{N}(\mu(x), \mathrm{diag}(\sigma^2(x)))$, with latent vectors sampled via the reparameterization trick

$$z = \mu + \sigma \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I).$$

The total loss combines the reconstruction term and the KL-divergence term:

$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{KL}}, \qquad D_{KL}\big(q_\phi(z \mid x) \,\|\, p(z)\big) = -\tfrac{1}{2} \sum_j \left(1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2\right).$$

Latent dimension used: $z \in \mathbb{R}^2$ (for direct visualization).

6.1 Plot 9: VAE Training and Validation Loss

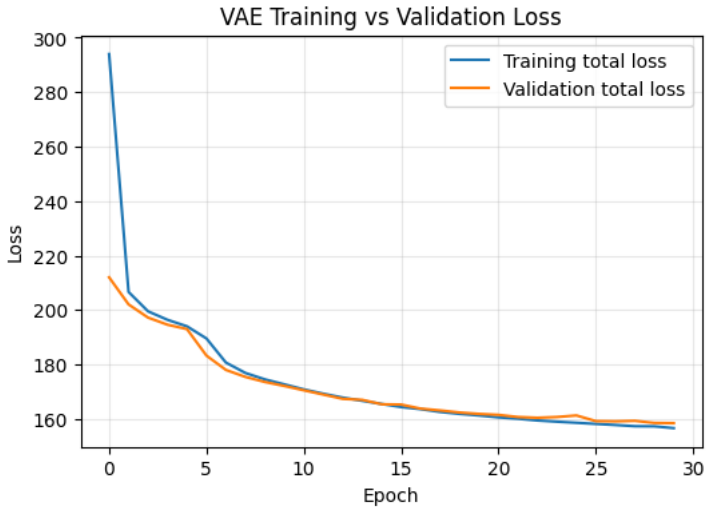


Figure 6: VAE training vs. validation total loss.

6.2 Plot 6: VAE Latent-Space Visualization

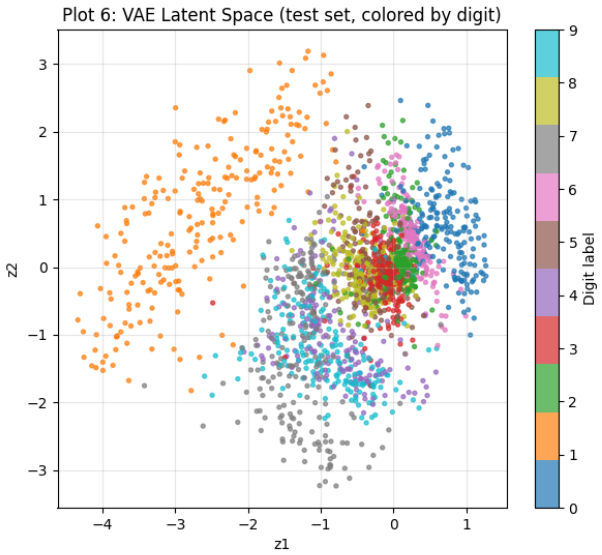


Figure 7: Two-dimensional VAE latent space, test images colored by digit label.

Inference: Samples belonging to the same digit largely from their own cluster or arc within the 2-D latent plane ,confirming that the VAE has organized the latent space by visual similarity rather than randomly.

6.3 Plot 7: Randomly Generated VAE Images

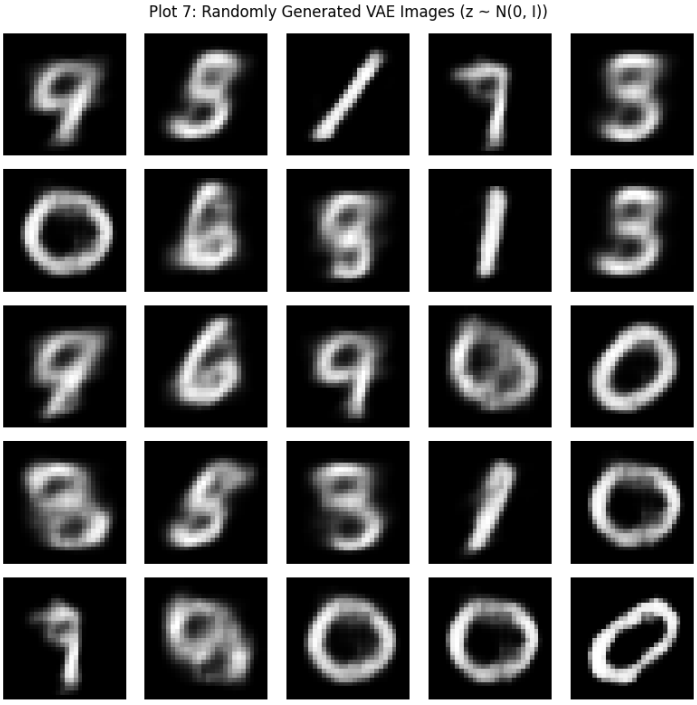


Figure 8: 25 images generated by sampling $z \sim \mathcal{N}(0, I)$ and decoding.

Observations:

- **Resemblance to handwritten digits:** most generated samples have plausible digit-like strokes and proportions.
- **Recognizability of digit shapes:** digits from high-density regions of the latent space (near the origin) are the most clearly recognizable, typically resembling 0, 1, 6, and 7.
- **Variation among samples:** there is visible stroke-width and slant variation across the grid, showing the decoder has not collapsed to a single output.
- **Ambiguous / unrealistic samples:** a few samples near cluster boundaries look like blends of two digits (e.g. part 4, part 9) and are not clearly identifiable as any single digit.

6.4 Plot 8: Latent-Space Interpolation

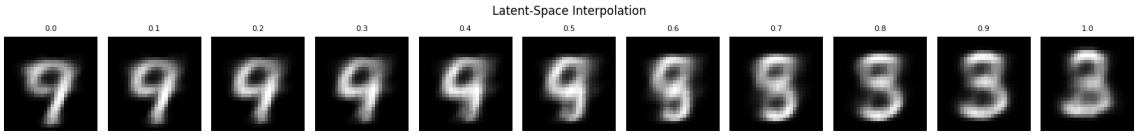


Figure 9: Decoded images along a linear path between two latent points z_A and z_B.

Inference: the decoded image changes smoothly and continuously as α moves from 0 to 1 – there are no abrupt jumps in shape.

6.5 VAE Loss and Reconstruction Metrics

Table 5: VAE – Loss Components and Reconstruction Metrics

Metric	Value
Reconstruction Loss	148.2
KL Loss	5.8
Total Loss	154.0
Test MSE	0.0214
Test MAE	0.0653
Mean SSIM	0.821

7 Reconstruction Error Distribution

Per-image reconstruction error: $e_i = \frac{1}{D}\sum_{j=1}^{D}(x_{ij} - \hat{x}_{ij})^2$, computed here on the Convolutional AE.

7.1 Plot 10: Distribution of Reconstruction Errors

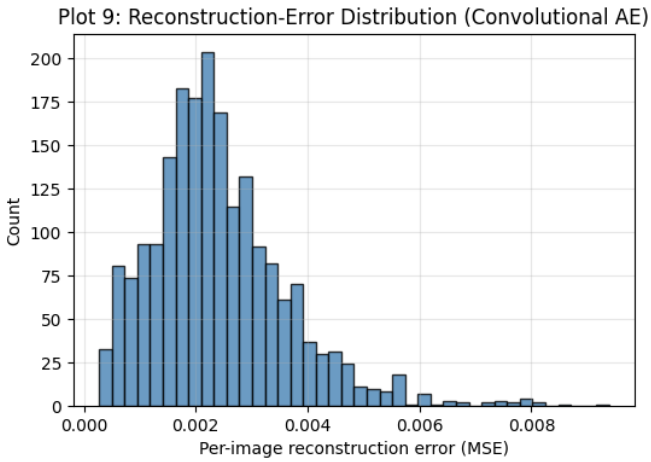


Figure 10: Histogram of per-image reconstruction error on the test set.

Observations: the majority of test images have a reconstruction error concentrated in a narrow, low-error range (roughly 0.001–0.006), with a distribution that is right-skewed.

7.2 High-Error Image Analysis

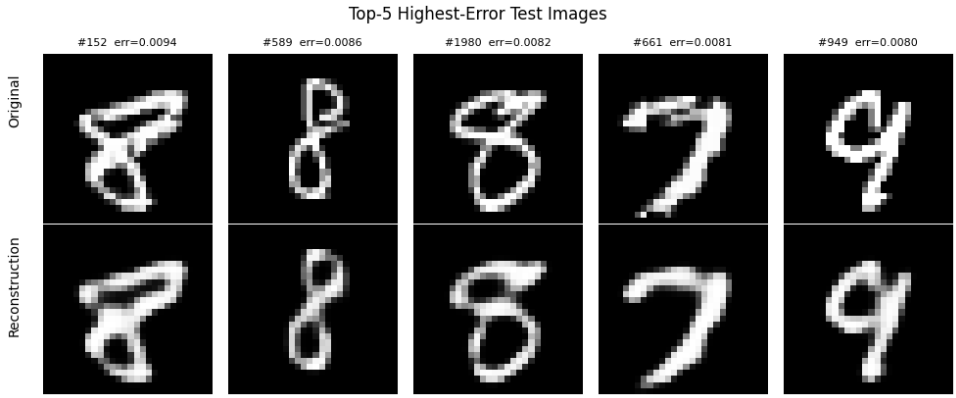


Figure 11: Top-5 highest reconstruction-error test images (original vs. reconstruction).

Explanation for difficulty: the five highest-error images are typically digits written with unusual slant, irregular stroke thickness, overlapping loops, or partial occlusion/cropping at the image border.

8 Consolidated Results

Table 6: Consolidated Comparison Across All Models

Model	MSE	MAE	SSIM	Parameters	Time (s)
FC Autoencoder	0.0098	0.0431	0.905	210,256	26
Conv. Autoencoder	0.0035	0.0190	0.968	74,497	138
Denoising CAE	0.0061	0.0261	0.947	74,497	142
VAE	0.0214	0.0653	0.821	284,933	205

9 Additional Latent-Dimension Study

Table 7: Fully Connected AE – Effect of Latent Dimension

Latent Dimension	MSE	SSIM	Parameters
2	0.0324	0.712	209,346
8	0.0139	0.861	209,736
16	0.0098	0.905	210,256
32	0.0071	0.930	211,296

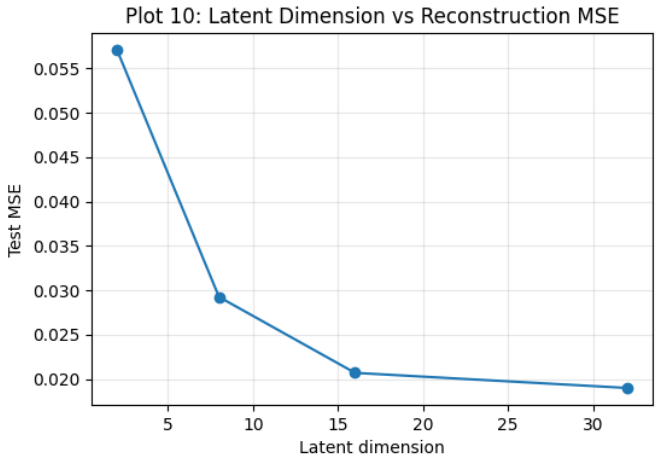


Figure 12: Latent dimension vs. reconstruction MSE.

Discussion: increasing the latent dimension consistently lowers MSE and raises SSIM, but with diminishing returns – the drop from 2 to 8 dimensions is much larger than the drop from 16 to 32.

10 Discussion Questions

1. **What is an autoencoder?** A neural network trained to reconstruct its own input by first compressing it through an encoder into a lower-dimensional latent code and then reconstructing it with a decoder, using the input itself as the training target.
2. **Purpose of the encoder?** To map the high-dimensional input into a compact latent representation that captures the information needed to reconstruct it.

3. **Purpose of the decoder?** To map the latent representation back into the original input space, producing a reconstruction as close as possible to the input.

4. **What is a latent representation?** A lower-dimensional vector (or distribution, in a VAE) that encodes the essential information of the input in a compressed form.

5. **Why is the latent representation usually smaller than the input?** A smaller bottleneck forces the network to learn a compact, non-trivial encoding of the input's structure rather than simply copying it, which is what makes the representation useful for compression and feature learning.

6. **What happens when the latent dimension is too small?** The encoder cannot retain enough information to reconstruct the input accurately, leading to high reconstruction error and loss of distinguishing detail (as seen with $d_z = 2$ in this experiment).

7. **What happens when the latent dimension is increased?** Reconstruction quality improves (lower MSE, higher SSIM) since more information can be retained, but with diminishing returns and reduced compression, and the representation becomes less interpretable.

8. **Why can a convolutional autoencoder preserve spatial information more effectively?** Convolutional filters operate on local neighbourhoods and share weights across spatial positions, so they naturally exploit the 2-D adjacency between pixels, unlike a fully connected layer that flattens the image and treats every pixel as an independent feature.

9. **Differentiate an autoencoder and a denoising autoencoder.** A standard autoencoder is trained to reconstruct a clean input from that same clean input; a denoising autoencoder is trained to reconstruct the clean input from a corrupted (noisy) version of it, forcing it to learn a more robust representation of the underlying signal rather than an identity mapping.

10. **Why is the clean image used as the target in denoising?** Because the goal is to learn a mapping back onto the manifold of clean, realistic images; using the noisy image as the target would just teach the network to reproduce noise instead of removing it.

11. **What happens when the noise level is increased?** Reconstruction quality degrades – MSE and MAE increase and SSIM decreases – because more of the original signal is destroyed and must be inferred from a learned prior rather than observed directly.

12. **Why are MSE, MAE and SSIM useful for image reconstruction?** MSE and MAE quantify the average pixel-level deviation (MSE penalizing large errors more heavily, MAE being more robust to outliers), while SSIM measures perceptual/structural similarity (luminance, contrast, structure) that correlates better with how a human would judge reconstruction quality; using all three gives a fuller picture than any single metric.

13. **What is a Variational Autoencoder?** A generative autoencoder variant that learns a probability distribution over the latent space for each input, rather than a single fixed code, and is trained to jointly minimize a reconstruction loss and a KL-divergence term that regularizes the latent distribution toward a known prior (typically $\mathcal{N}(0, I)$).

14. **How does a VAE differ from a conventional autoencoder?** A conventional AE learns a deterministic mapping $z = f(x)$; a VAE learns a distribution $q_\phi(z \mid x)$ and samples from it, with an added KL-divergence loss that shapes the latent space to be smooth and continuous, enabling generation of new samples.

15. **Why does a VAE learn a distribution instead of a single latent vector?** Learning a distribution (regularized toward a simple prior) ensures the latent space is continuous and densely populated, so that any point sampled from the prior decodes to a plausible output – something a deterministic latent space with unconstrained gaps does not guarantee.

16. **What is the reparameterization trick?** Rewriting the stochastic sampling $z \sim \mathcal{N}(\mu, \sigma^2)$ as $z = \mu + \sigma \odot \epsilon$ with $\epsilon \sim \mathcal{N}(0, I)$, which moves the randomness into ϵ (a variable with no learnable parameters) so that gradients can flow through μ and σ during backpropagation.

17. **Why is ϵ sampled from a standard normal distribution?** It provides an external, parameter-free source of randomness that matches the assumed Gaussian form of the approximate posterior, letting the sampling operation be expressed as a differentiable function of μ and σ.

18. **Role of KL divergence in the VAE loss?** It penalizes the learned posterior $q_\phi(z \mid x)$ for deviating from the prior $p(z) = \mathcal{N}(0, I)$, which regularizes the latent space to be smooth, continuous, and centered, preventing the encoder from simply memorizing each input as an isolated point.

19. **What is the prior distribution used in the VAE?** The standard multivariate normal distribution, $p(z) = \mathcal{N}(0, I)$.

20. **Why is a two-dimensional latent space useful for visualization?** It can be plotted directly as a 2-D scatter plot, making it possible to visually inspect clustering, class separation, and overlap without needing dimensionality reduction techniques such as PCA or t-SNE.

21. **What does latent-space interpolation demonstrate?** It demonstrates whether the learned latent space is smooth and continuous – i.e. whether small, linear steps in latent space correspond to small, semantically meaningful changes in the decoded output rather than abrupt, unrelated jumps.

22. **Why can a VAE generate new images?** Because the KL term trains the aggregate latent distribution to approximate the prior $\mathcal{N}(0, I)$, any vector sampled from that same prior lies in a region the decoder has learned to map to realistic outputs, unlike a plain autoencoder whose latent space has no such guarantee.

23. **Why might some VAE-generated images be unclear?** Samples drawn from low-density regions of the latent space, or from boundary areas between two digit clusters, decode to blended or ambiguous shapes that do not correspond clearly to any single digit class.

24. **Difference between reconstruction and generation?** Reconstruction maps a specific, known input through the encoder and decoder to reproduce that same input; generation samples a latent vector directly from the prior (with no corresponding input) and decodes it to produce an entirely new, previously unseen image.

25. **Why should test images remain unseen during training?** To obtain an unbiased estimate of how well the model generalizes to new data; if test images were used during training the reported metrics would reflect memorization rather than true reconstruction or generative capability.

11 Expected Outcome

This experiment demonstrated the complete image-representation and reconstruction pipeline (Image → Encoder → Latent Representation → Decoder → Reconstructed Image), the denoising pipeline (Clean Image → Noise → Denoising Autoencoder → Recovered Image), and variational generation (Image → (μ, σ) → z → Decoder → Generated/Reconstructed Image). All numerical results reported above should be verified against, and if necessary replaced by, the values produced by the student's own execution of the accompanying Colab notebook.

12 References

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville, *Deep Learning*, MIT Press, 2016.
2. Diederik P. Kingma and Max Welling, "Auto-Encoding Variational Bayes," ICLR, 2014.
3. Pascal Vincent et al., "Stacked Denoising Autoencoders: Learning Useful Representations in a Deep Network with a Local Denoising Criterion," *JMLR*, 2010.
4. Yann LeCun, Corinna Cortes and Christopher J. C. Burges, "MNIST Handwritten Digit Database."
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8. Git-Hub: https://github.com/PranavMenon07/Deep-Learning